

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis

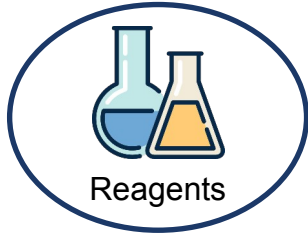


Background models

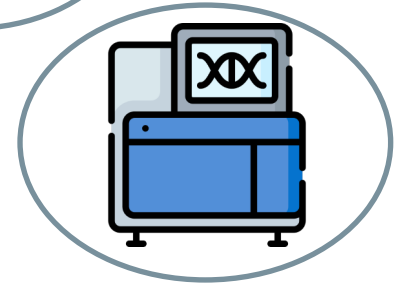
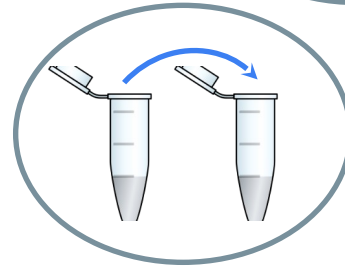
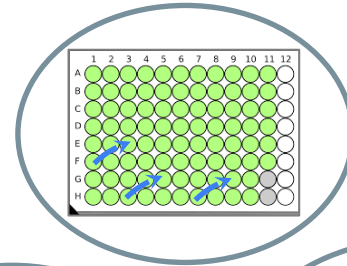
Exploring Results

- QC metrics
- Sample Report
 - Metrics
 - Analysis
- Accounting for background

Contamination will happen!



External
Contamination

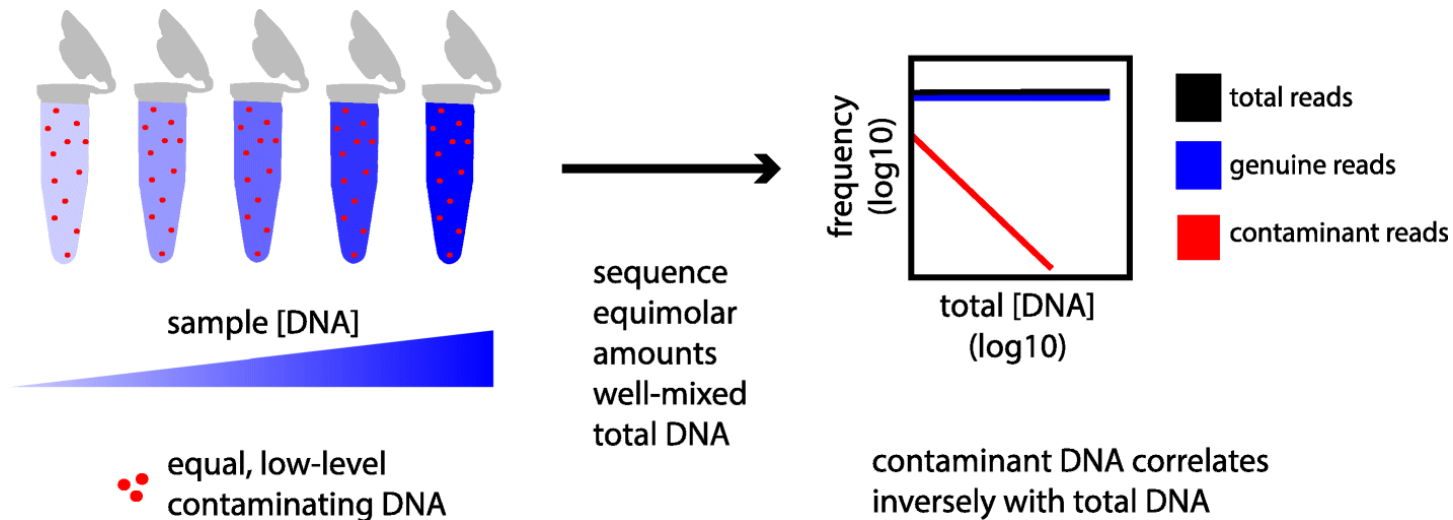


Internal
Contamination
(cross-contamination)

Contamination can lead to inaccurate data interpretation

- Falsely inflates within-sample diversity
- Obscures differences between samples
- Interferes with comparisons across studies

Contamination disproportionately affects low-biomass samples



Water controls can help

- 'Blanks' such as water controls can help identify both internal and external contaminants
- Water controls should be processed the same way as the samples, using the same reagents
 - Have at least 2 water controls, but the more the better!

Background model

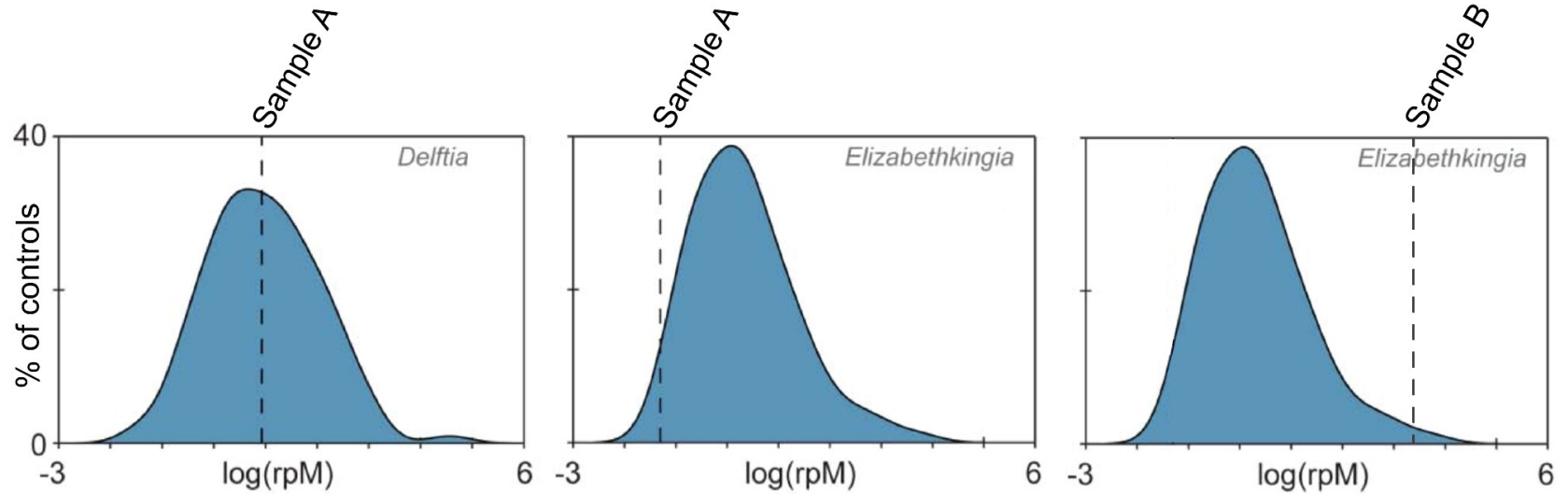
- Used to calculate relative abundance of the taxa found in negative controls
- Background model based on relative abundance metric:
 - Uses the bases per million (bpm) as the metric of relative abundance.

Background model enables Z-score calculation

- **Z-score**

- Measurement that describes a value's relationship to the mean of a group of values
 - How does the abundance of a taxon in a sample compare to its mean abundance in negative controls?

Background distribution
overlaid with sample taxa abundance



----- **Abundance in potentially infectious sample**

 **Distribution of abundance in controls**

Background model enables Z-score calculation

- **Z-score**

- Measurement that describes a value's relationship to the mean of a group of values
 - How does the abundance of a taxon in a sample compare to its mean abundance in negative controls?
- It is measured in terms of standard deviations from the mean

$$Z = \frac{x - \mu}{\sigma}$$

$$Z = \frac{x - \mu}{\sigma}$$

x = taxon abundance in sample

μ = taxon mean abundance in negative controls
(background model)

σ = standard deviation of taxon abundance in negative
controls (background model)

Z-score linear scale:



0

$$Z = \frac{x - \mu}{\sigma}$$

x = taxon abundance in sample

μ = taxon mean abundance in negative controls
(background model)

σ = standard deviation of taxon abundance in negative
controls (background model)

No difference

$$Z = 0$$


$$Z = \frac{x - \mu}{\sigma}$$

x = taxon abundance in sample

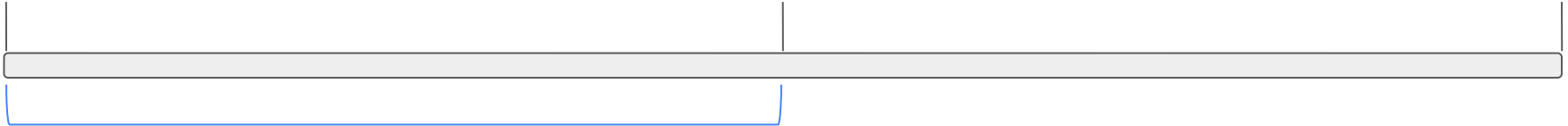
μ = taxon mean abundance in negative controls
(background model)

σ = standard deviation of taxon abundance in negative
controls (background model)

No difference

$Z < 0$

$Z = 0$



Taxa **more abundant in negative controls** than
in samples

$$Z = \frac{x - \mu}{\sigma}$$

x = taxon abundance in sample

μ = taxon mean abundance in negative controls
(background model)

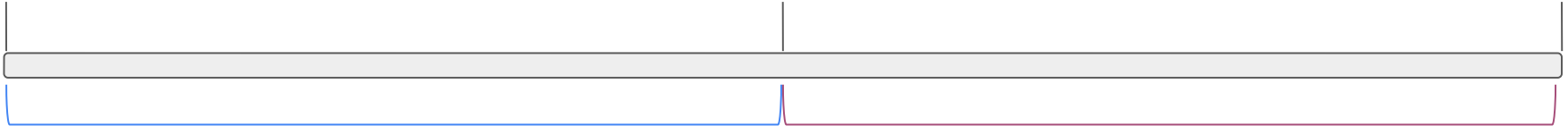
σ = standard deviation of taxon abundance in negative
controls (background model)

No difference

$Z < 0$

$Z = 0$

$Z > 0$



Taxa **more abundant in negative controls** than
in samples

Taxa **more abundant in samples** than negative
controls

$$Z = \frac{x - \mu}{\sigma}$$

x = taxon abundance in sample

μ = taxon mean abundance in negative controls
(background model)

σ = standard deviation of taxon abundance in negative
controls (background model)

Only in
control
-100

No difference

0

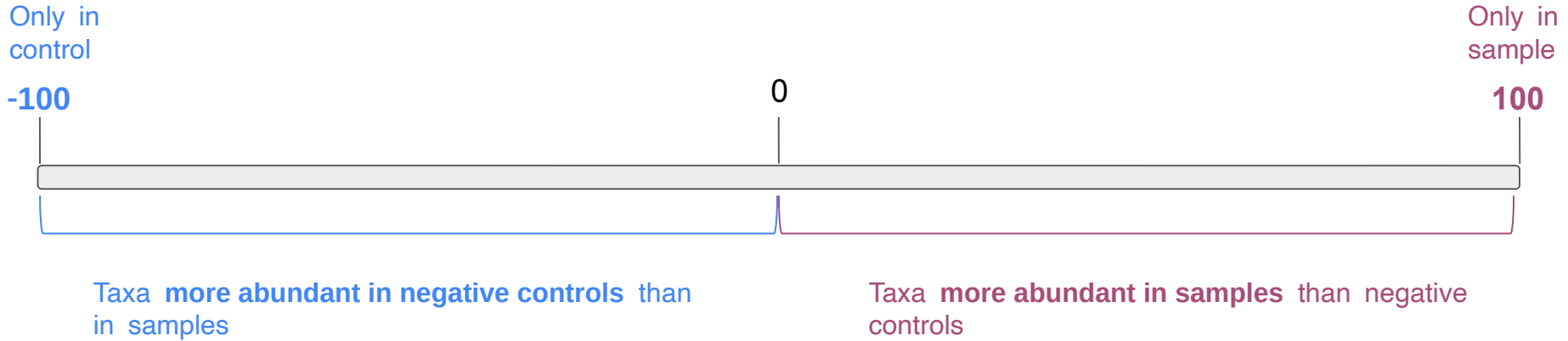
Only in
sample
100

Taxa **more abundant in negative controls** than
in samples

Taxa **more abundant in samples** than negative
controls

Use Z-score filter after applying background model

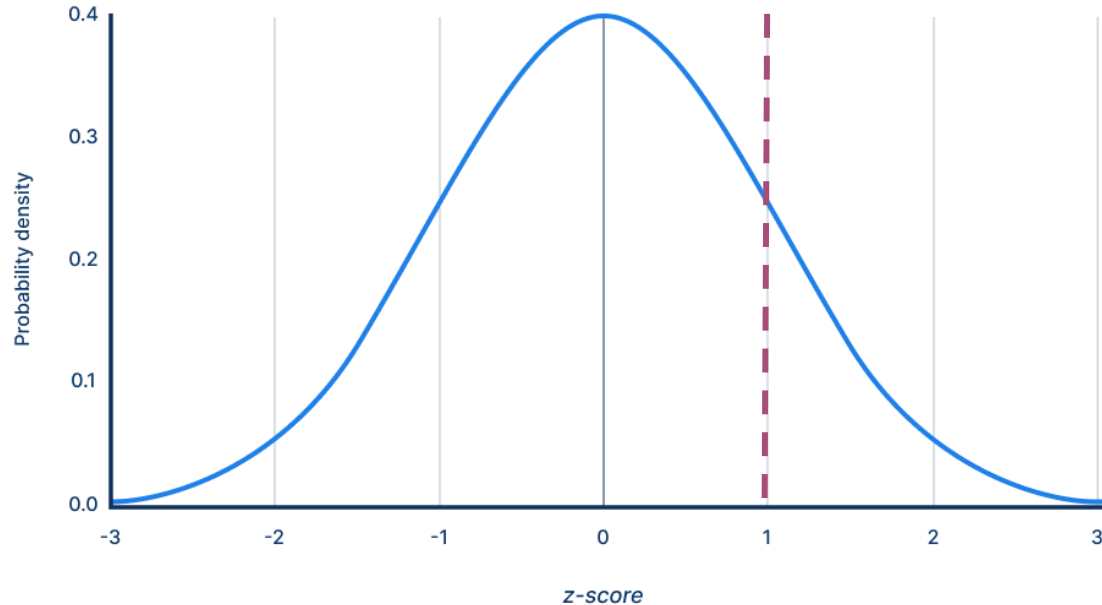
Start with taxa with Z-scores > 1



Use Z-score filter after applying background model

Start with taxa with Z-scores > 1

Standard normal distribution



Calculating Z-score

Standard Z-score calculation for sample “S”:

$$Z = \frac{(\text{Taxon rPM in S}) - (\text{Taxon mean rPM in Negative Controls})}{\text{Standard deviation for Taxon rPM in Negative Controls}}$$

Additional controls

- Controls can also be used to create a background model to identify pathogens of interest (for example, cases vs controls)
- Sample controls can also be used to identify contaminants from sampling, but can be difficult to come by
- Background models can be applied to the sample report and the heatmap

Let's make a background model!

Option 1 (outlined in worksheet): simple, subtract average signal in water controls from signal found in sample

Option 2 (shown in this slide deck): calculate a Z score for each taxa within each sample, using the equation outlined above

Go to reports and look at the water controls

- Are your candidate taxa present in the water controls?
- If so, at what level?
- How does this compare to the level seen in your sample?

→ Subtract the background / water control signal from the sample signal for each taxa



Interpret: real signal should be higher in samples than in water controls

Example

Example: serum sample from VHF-suspect samples

RR145011_hac ▾ **Suspected sample**

[Share](#) [Download ▾](#) [...](#)

[Nanopore](#)

Taxon name

Name Type: Scientific ▾

Categories: **1** ▾

Threshold filters ▾

Read Specificity: Specific Only ▾

Annotation ▾

Pathogen Flags ▾

Viruses X

4 rows passing the above filters, out of 120 total rows [CLEAR FILTERS](#)



▾	Taxon	bPM ▾	b	r	contig	cont...	%id	L	E val...	NT NR
▾ ▢	Orthoflavivirus (1 viral species: ● 1)	14,096.0 0.0	56,794 0	62 0	0 0	0 0	96.5 0.0	754.7 0.0	10 ⁻²⁸⁶ 0	
	▢ Orthoflavivirus denguei Known Pathogen	14,096.0 0.0	56,794 0	62 0	0 0	0 0	96.5 0.0	754.7 0.0	10 ⁻²⁸⁶ 0	
▾ ▢	Lentivirus (1 viral species)	6,503.7 0.0	26,204 0	42 0	0 0	0 0	90.2 0.0	603.0 0.0	10 ⁻²⁴⁷ 0	
	▢ Human immunodeficiency virus 1	6,503.7 0.0	26,204 0	42 0	0 0	0 0	90.2 0.0	603.0 0.0	10 ⁻²⁴⁷ 0	

Example: serum sample from VHF-suspect samples

Neg_lib_ctrl2_hac ▾

Water control

Share

Download ▾

⋮

Nanopore

Taxon name

Q

Name Type: Scientific ▾

Categories: 1 ▾

Threshold filters ▾

Read Specificity: Specific Only ▾

Annotation ▾

Pathogen Flags ▾

Viruses X

2 rows passing the above filters, out of 77 total rows

CLEAR FILTERS

⌵

Taxon

bPM ▾

b

r

contig

cont...

%id

L

E val...

NT
NR

▾ ▢ Orthoflavivirus (1 viral species: ● 1)

19,932.9
0.0

10,850
0

12
0

0
0

0
0

96.8
0.0

758.1
0.0

10⁻²⁸⁸
0

▢ Orthoflavivirus denguei

Known Pathogen

19,932.9
0.0

10,850
0

12
0

0
0

0
0

96.8
0.0

758.1
0.0

10⁻²⁸⁸
0

Example: serum sample from VHF-suspect samples

Suspected sample

bPM ∨	b	r
14,096.0	56,794	62
0.0	0	0
14,096.0	56,794	62
0.0	0	0

Water control

bPM ∨	b	r
19,932.9	10,850	12
0.0	0	0
19,932.9	10,850	12
0.0	0	0

Serum library has high signal from barcode bleeding

		Sample		Water		Sample - water		
sample_name	Virus	NT bPM	NT r	NT bPM water	NT r water	bPM - water	r - water	Conclusion
RR145011	Dengue	14096	62	19932	12	-5836	50	Background
	HIV	6503	42	-	-	6503	42	Hit

		Sample		Water		Sample - water		
sample_name	Virus	NT bPM	NT r	NT bPM water	NT r water	bPM - water	r - water	Conclusion
RR145011	Dengue	14096	62	19932	12	-5836	50	Background
	HIV	6503	42	-	-	6503	42	Hit

Subtracting background signal was necessary to help with interpretation and reduce false positives

Example report - background subtraction necessary for accurate calls

RR145011_hac ▾

Nanopore

Taxon name Name Type: Scientific ▾ Categories: 1 ▾ Threshold filters ▾ Read Specificity: Specific Only

Viruses X

4 rows passing the above filters, out of 120 total rows [CLEAR FILTERS](#)

▼	Taxon	bPM ▾	b	r	contig	cont...	%id	L	E val...	NT NR
▼ ▢	Orthoflavivirus (1 viral species: ● 1)	14,096.0 0.0	56,794 0	62 0	0 0	0 0	96.5 0.0	754.7 0.0	10 ⁻²⁸⁶ 0	
	▢ Orthoflavivirus denguei Known Pathogen	14,096.0 0.0	56,794 0	62 0	0 0	0 0	96.5 0.0	754.7 0.0	10 ⁻²⁸⁶ 0	
▼ ▢	Lentivirus (1 viral species)	6,503.7 0.0	26,204 0	42 0	0 0	0 0	90.2 0.0	603.0 0.0	10 ⁻²⁴⁷ 0	
	▢ Human immunodeficiency virus 1	6,503.7 0.0	26,204 0	42 0	0 0	0 0	90.2 0.0	603.0 0.0	10 ⁻²⁴⁷ 0	

This DENV is
background /
barcode hopping

Subtracting background signal was necessary for some samples to help with interpretation and reduce false positives

Analysis Workflow - ONT

Complete a first pass sample report view

1. Look at the list of detected taxa
2. Record NT bPM, NT r, and any other relevant metrics
3. Look at the coverage plot for each taxa and record your interpretation (low, medium, high coverage)
4. BLASTn contigs from each candidate taxa and record what you find (specific, non-specific, etc)

Check negative water controls

1. Navigate to the sample report for each negative water control included in the run
2. For each taxa observed in the samples, record the NT bPM, NT r, and any other metrics in each water control
3. If you have >1 water control, calculate the average signal for each taxa

Put it all together (background subtraction + interpretation)

1. Subtract the background/water control signal from the sample signal for each taxa
2. Interpret: real signal should be higher in samples than in water controls

Analysis Workflow - ONT

Complete a first pass sample report view

1. Look at the list of detected taxa
2. Record NT bPM, NT r, and any other relevant metrics
3. Look at the coverage plot for each taxa and record your interpretation (low, medium, high coverage)
4. BLASTn contigs from each candidate taxa and record what you find (specific, non-specific, etc)

Check negative water controls

1. Navigate to the sample report for each negative water control included in the run
2. For each taxa observed in the samples, record the NT bPM, NT r, and any other metrics in each water control
3. If you have >1 water control, calculate the average signal for each taxa

Put it all together (background subtraction + interpretation)

1. Subtract the background/water control signal from the sample signal for each taxa
2. Interpret: real signal should be higher in samples than in water controls

Go to worksheet:
Identification of
background signal

Current Worksheet: CZ ID Training Reference Tables & Worksheets_ONT, pg 17-20

Next Slides: DL8-O_Heatmap App for ONT